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NLP-Based Prediction of Medical Specialties at Hospital Admission Using Triage Notes

Émilien Arnaud
Emergency Department
Amiens-Picardy University Hospital
Amiens, France
arnaud.emilien@chu-amiens.fr

Mahmoud Elbattah
Laboratoire MIS
Université de Picardie Jules Verne
Amiens, France
mahmoud.elbattah@u-picardie.fr

Maxime Gignon
Public Health Department
Amiens-Picardy University Hospital
Amiens, France
maxime.gignon@chu-amiens.fr

Gilles Dequen
Laboratoire MIS
Université de Picardie Jules Verne
Amiens, France
gilles.dequen@u-picardie.fr

***Abstract*— Data Analytics is rapidly expanding within the healthcare domain to help develop strategies for improving the quality of care and curbing costs as well. Natural Language Processing (NLP) solutions have received particular attention whereas a large part of clinical data is stockpiled into unstructured physician or nursing notes. In this respect, we attempt to employ NLP to provide an early prediction of the medical specialties at hospital admission. The study uses a large-scale dataset including more than 260K ED records provided by the Amiens-Picardy University Hospital in France. Our approach aims to integrate structured data with unstructured textual notes recorded at the triage stage. On one hand, a standard MLP model is used against the typical set of features. On the other hand, a Convolutional Neural Network is used to operate over the textual data. While both learning components are conducted independently in parallel. The empirical results demonstrated a promising accuracy in general. It is conceived that the study could be an additional contribution to the mounting efforts of applying NLP methods in the healthcare domain.**

Keywords—NLP, Deep Learning, Hospitalization, Emergency Department.

I. INTRODUCTION

A long-standing challenge for hospitals is the overcrowding experienced in the Emergency Department (ED) units. Faced by continuously increasing demand, the ED crowding has become an international issue throughout the hospitals worldwide. Plentiful studies, from different countries, emphasized that crowded EDs could lead to unintended breaches and deficiencies in healthcare [1]. Overcrowding could generally have an adverse impact on several aspects pertaining to the quality and cost of healthcare. More specifically, examples include increased boarding time [2], higher mortality rates (e.g. [3] [4]), and increased morbidity [5]. Accordingly, there is a dire need for developing utilities that can assist with the management and administration of ED services.

In this respect, Data Analytics has been considered as part of the solution. Healthcare is delivered in data-rich environments where abundant procedures generate vast amounts of data continuously. Likewise, the triage stage typically includes multiple readings and observations, frequently made by nurses and physicians during the ED reception of patients. Therefore, Machine Learning (ML) is generally well-poised to serve as an enabling artefact for improving multiple aspects of the ED operations. While Text Analytics is particularly beneficial in this regard since a large part of the triage data is typically stored into unstructured free-text notes.

In this context, this study is concerned with applying Natural Language Processing (NLP) methods for predicting medical specialties at an early stage of hospital admission. The study is based on a large-scale dataset collected by the Amiens-Picardy University Hospital in France. The dataset contained about 260K ED records collected over more than four years. In essence, our approach aims at integrating structured data with unstructured textual notes taken by nurses or physicians at the triage stage.

The study attempts to make a set of contributions as follows. First, the study practically explores the suitability of free-text triage notes for developing ML models for predicting medical specialties. We present a Deep Learning-based architecture that could integrate structured data along with unstructured text to this end. Perhaps equally important, the study could be regarded as an exemplary case of using triage notes in the French language, whereas the literature still generally lacks studies using languages other than English. In a broader sense, the study is hoped to contribute to the endeavors towards leveraging unstructured textual data to support the operations of healthcare. The study is aimed to be an extension of our earlier work [6], which focused on predicting patient hospitalization.

II. RELATED WORK

The literature is replete with studies that sought to avail of Text Analytics methods in the healthcare context. In this regard, there have been plentiful attempts in order to utilize unstructured data such as free-text notes made by nurses or physicians. This section aims to review representative contributions that implemented Text Analytics in a diversity of applications to support the clinical decision-making.

A study developed a Text-Analytics solution for the automatic detection of medical events using Electronic Health Record (EHR) data [7]. The textual records included data collected from three University hospitals based in France over the period October 2009 to December 2010. The dataset spanned a variety of surgical specialties such as neurosurgery, orthopedic surgery, and digestive surgery. The system performance was

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compared with other standard methods. The system could achieve about 84% for both of the sensitivity and specificity. The study largely confirmed the feasibility of using NLP-based methods to automate the detection and monitoring of healthcare-associated events in hospital facilities. By the same token, NLP methods were applied for the automatic identification of Critical limb ischemia (CLI) [8]. The dataset included narrative clinical notes retrieved from the EHR database. The model performance was validated against the human-based abstraction of clinical notes. Specifically, a physician reviewed and interpreted the information in the EHR data for each patient in the dataset. Overall, the method could achieve an F1-score up to about 90%.

While other studies explored the potentials of Text Analytics to avail of the textual data in medical prescriptions. For instance, an NLP-based approach was developed for the automatic classification of medical prescriptions [9]. They could develop an effective classifier based on the data pertaining to patient/doctor personal data, symptoms, pathology, diagnosis, and suggested treatments. Their results reported that only 5% of the prescriptions could not be automatically classified. Furthermore, part of the recent efforts explored the applicability of NLP methods to predict the International Classification of Diseases (ICD) codes. The manual encoding process is usually time-consuming, and prone to various errors as well. For instance, a study applied topic mining and Deep Learning to automatically predict the ICD codes from free-text medical records [10]. They used the MIMIC-III dataset, which provides a large freely accessible repository of ICU records. The reported results indicated that their method could increase the F1-score approximately by 5% compared to the previous work.

While many other studies sought to develop predictive models to help streamline the process of hospital admissions. Increasing contributions attempted to utilize unstructured data such as free-text notes made by nurses or physicians at ED units. For instance, a study [11] utilized the bag-of-words approach to extract features from free-text notes recorded at triage. Using a dataset of over 250K ED visits, neural network models were trained to predict hospital admissions. They could achieve a promising accuracy of about 74%. While another study [12] aimed to compare the performance of ML models with the inclusion of textual elements. Deep Learning was applied along with Word Embeddings using clinical narratives. They demonstrated that the model accuracy generally improved with the addition of free-text fields.

More recent studies applied the state-of-the-art implementations of NLP with the Bidirectional Encoder Representations from Transformers (BERT) [13]. The BERT approach brings the advantage of using pre-trained models to tackle a broad set of tasks. In this regard, a framework was developed for transforming free-text descriptions into a standardized form based on the Health Level 7 (HL7) standards [14]. They utilized a combination of domain-specific knowledgebases along with BERT models. It was demonstrated that the BERT-based language representation contributed significantly to the model performance. The literature includes further contributions that made use of the BERT approach for a variety of Text Analytics tasks (e.g. [15] [16]).

III. METHODOLOGY

A. Approach Overview

Our approach aims to apply a multi-input of mixed data including standard features (e.g. numeric and categorical) along with textual data from triage notes. On one hand, a standard Multi-Layer Perceptron (MLP) model is used with the standard set of features. On the other hand, a Convolutional Neural Network (CNN) is used to operate over the textual data to perform the feature extraction from text. The two components of learning are conducted independently in parallel during the training process. As such, our approach is based on applying a dual channel of learning simultaneously.

Subsequently, a global feature map is constructed by concatenating the outputs produced by the MLP and CNN modules. The feature map is eventually fed to another MLP module, which performs the classification task. Figure 1 sketches the key elements of our approach.

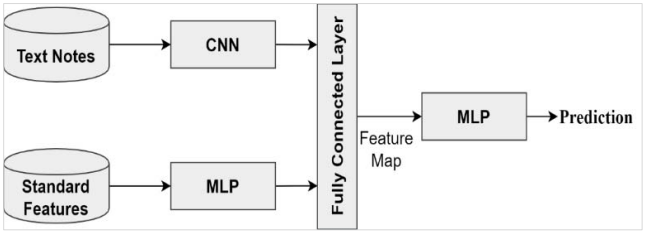


Fig 1. Approach overview.

B. Data Desciption

The present work used a dataset provided by the Amiens-Picardy University Hospital (APUH) in France. The authors had obtained all permissions required for accessing the dataset. More specifically, the study was approved by the "Hors Loi Jarde Committee" under the reference PI2019_843_0066. Moreover, all personal information was fully anonymized in order to maintain the privacy of patients.

The dataset contained more than 260K ED records spanning more than four years (January 2015-June 2019). Each record was associated with a binary label representing the outcome at triage (i.e. hospitalization or discharge). The percentage of hospitalized patients was about 35%. The average length of stay (LOS) in ED was 4h23m. While the average LOS of discharged patients was 4h04m, which was about 18% lower than those hospitalized. Table 1 gives summary statistics of the dataset.

TABLE 1. SUMMARY STATISTICS.

Year	Count of Records	Patient Hospitalization %
2019	30,615	33.1%
2018	61,794	34.0%
2017	60,155	34.1%
2016	57,963	35.4%
2015	53,465	35.6%

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TABLE 2. DATASET FIELDS.

#	Field Name	Type
1	Arrival (Week Day /Hour)	Categorical
2	Gender	
3	Origin	
4	Arrival Modlaity	
5	Accompaniers	
6	Family Status	
7	Waiting Modality	
8	Reason for Encounter	
9	Circumstances	
10	Age	Numeric
11	Oxygen Flow	
12	Heart Rate	
13	Respiration Rate	
14	Systolic Blood Pressure	
15	Diastolic Blood Pressure	
16	Pain Scale	
17	Temperature	
18	Oxygen Saturation	
19	Capillary Blood Glucose	
20	Capillary Blood Hemoglobin	
21	Bladder Volume	
22	Capillary Blood Ketones	
23	Breath Test of Alcohol	
24	Nurse Triage Scale	
25	Nurse Notes	Text
26	Psychiatric History	
27	Surgical History	
28	Medical History	

The dataset included a variety of variables representing numeric, categorical, and textual notes. Part of the numeric variables described standard vital signs such as temperature, heart rate, oxygen flow, respiratory rate, etc. While the categorical variables described general information about patients such as gender, origin, family status, etc. The categorical variables were defined based on data dictionaries provided by the hospital system. In addition, the dataset contained free-text data fields. Specifically, a set of four textual fields described the triage-provided observations, surgical history, psychiatric history, and medical history of patients. The free-text notes were input in the French language by nurses or physicians at the ED.

It should also be mentioned that the dataset suffered from blank textual fields across thousands of records. Precisely, about 70K records did not include any textual information at all. At the outset, a number of irrelevant variables were excluded from the raw dataset (e.g. Record ID, discharge date). The dataset eventually contained a set of 9 categorical variables, 15 numeric variables, and 4 text fields, as listed above in Table 2.

As well, the data records indicated the specialty assigned to hospitalized patients. The majority of specialties can be grouped into three broad categories including short-term hospitalization unit, surgery, or medical specialties (e.g. cardiology, neurology, etc.). Table 3 provides the proportions of specialties among hospitalized patients in the dataset.

TABLE 3. SUMMARY OF MEDICAL SPECIALTIES.

Specialty / Label	Hospitalization %
Surgery / CHIR	19.7%
Short-Term Hospitalization Unit / UHCD	42.4%
Medical Specialty / MED	33%
Other	4.9%

C. Data Preprocessing

The pre-processing of data is a challenging part due to the mixed-input approach, which required different types and order of procedures. In this regard, different pipelines of pre-processing were applied on each side of the input data. This section only describes the pre-processing procedures of numeric and categorical features.

Several columns of the dataset originally included missing values. The missing values of vital signs (e.g. blood pressure, temperature, etc.) were filled with routine values commonly given in the medical domain. Blank categorical values were set as per default values defined by the data dictionaries. The min-max normalization was applied to all numeric variables, such that all values were linearly scaled to [0, 1]. While the categorical features were encoded as one-hot numeric arrays. The pre-processing procedures were implemented using Python libraries including Pandas [17] and Scikit-Learn [18].

D. Text Normalization

As mentioned earlier, the dataset contained four columns, which stored the triage notes (see Table 2). Initially, we combined those columns into a single text field. The merged text can be regarded as a unique document associated with each patient record. Subsequently, standard procedures of text normalization were applied to clean and standardize the textual data into an amenable representation for developing ML models.

The initial procedures of normalization included case conversion, removing stop words, and spell checking. Subsequently, word stemming was applied to obtain the base form of words. Stemming generally helps to simplify the representation of text. The SnowballStemmer algorithm [19] was used. The normalization process was implemented using Python libraries including NLTK [20], and pyspellchecker [21] which supports spell checking for the French language.

E. Tokenization

Following the normalization process, the text was in a suitable form for tokenization. Text tokenization is defined as the process of splitting textual data into smaller meaningful components called 'tokens' [22].

We used the Keras library [23], which included helpful APIs for tokenization. Keras provides the Tokenizer utility class for transforming textual sentences into typical vectors, encoded as a list of integer values. Specifically, each integer is mapped to a value in a dictionary encoding the entire text corpus. While the vocabulary terms are represented by the dictionary keys.

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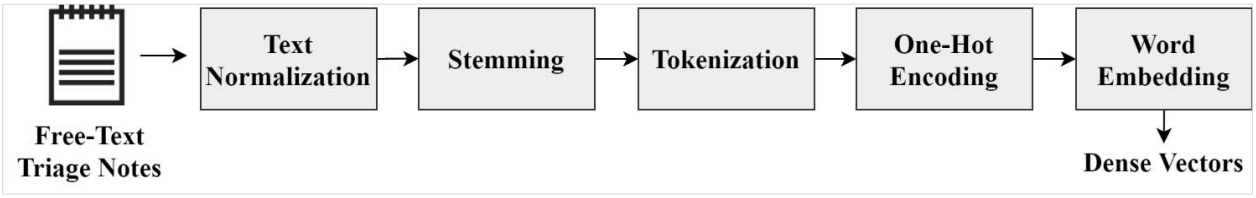


Fig 2. The pre-processing pipeline of textual data.

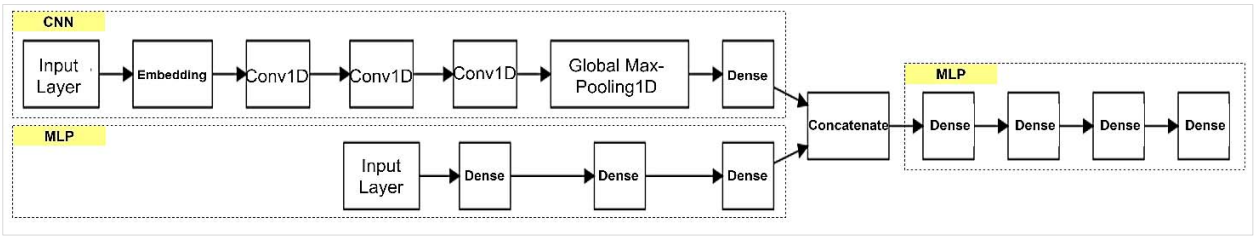


Fig 3. Model architecture.

F. *One-Hot Encoding*

Subsequently, the tokens were transformed into vectors by applying the one-hot encoding. The tokens (i.e. words) were represented by a vector of the vocabulary length. While Each word is assigned to an entry in the vocabulary, and the corresponding index is set to one. In this manner, the text sequences were eventually represented as binary vectors. The one-hot encoding scheme was implemented using the Keras APIs [23].

G. *Word Embedding*

The final step was to apply word embeddings, which is an important procedure to make the text sequences in a tractable form for training ML models. The output of one-hot encoding is usually very sparse vectors of high dimensionality. To this end, the embeddings were used to construct dense vectors of much lower dimensions.

In this regard, Keras [23] includes a special layer for implementing embeddings. The embedding layer basically serves as a dictionary that maps integer indices into dense vectors. The integer inputs are looked up in an internal dictionary, and then associated vectors are returned. The embedding layer was used as the top layer of the classification model. Figure 2 describes the pre-processing pipeline applied to transform the text all the way into dense vectors.

IV. EXPERIMENTS

As alluded earlier, the main task was to predict the medical specialty for patients admitted. The medical specialty was a multi-label classification task. An additional goal of the experiments was to explore the impact of including the textual data on the model performance.

A. *Experimental Set-up*

The process of data pre-processing and anonymization was conducted internally within the hospital computing facilities. This ensured the protection of the privacy of patient data.

Subsequently, the ML experiments were run on the MatriCS computing platform [24], hosted by the University of Picardy Jules Verne. The MatriCS platform provides sophisticated computing facilities including the state-of-the-art GPUs, typically needed for Deep Learning. Specifically, the Deep Learning models were implemented on a quad-GPU node comprising a set of four V-100 Nvidia GPUs.

B. *Model Design*

The model was designed based on a dual channel architecture, which allowed for dealing with the numeric and text data separately. On one channel, a standard MLP module was used to handle the numeric and categorical features. The MLP included three layers consisting of 128, 64, and 32 neurons, respectively.

The other channel included a CNN model to operate over the text data. One-dimensional convolutional layers were suitable to extract features from the text sequences. Specifically, the CNN model can be described as follows:

- A set of three 1-D convolutional layers, with kernel size=16, 32, and 64.
- A single pooling layer, applying a global-max operation.
- A single fully connected layer.

Subsequently, a global feature map was constructed by concatenating the outputs of the CNN and MLP channels. The feature map was eventually used for training another MLP module for performing the actual classification task. The MLP classifier included three hidden layers of 100, 50, and 25 neurons, respectively. Our experiments demonstrated that deeper architectures were not largely needed in our case. The ReLU activation function was used at all layers of the CNN and MLP modules. While the output layer used a SoftMax function, which was suitable for the multi-label classification task.

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In addition, the Dropout regularization approach [25] was used to help reduce over-fitting. The batch normalization [26] technique was also applied throughout the model layers. The classification models were implemented using Keras [23] with the TensorFlow backend [27]. An overview of the model architecture is sketched in Figure 3.

C. Specialty Classifier

As mentioned before, the prediction of specialty is a multi-label classification task. Specifically, the classifier was trained using a set of three labels including CHIR (i.e. surgery), UHCD (i.e. short-term hospitalization unit), and MED (i.e. medical specialty). The dataset was split into train and test sets based on a 3-fold cross-validation, while 20% of the train data was used for validation. However, the dataset only included the hospitalization records, which included specialty information. Figure 4 demonstrates the model loss in train and validation sets over 10 epochs with 20% of the train set used for validation.

Overall, the model could achieve about 68% accuracy of prediction. Equally important, the model performance was evaluated before and after including the text data in model training. In this regard, the sensitivity was calculated with respect to each label based on a one-against-all evaluation. It turned out that the model sensitivity declined by about 5% in case of excluding the text data. Specifically, the average sensitivity declined from 62% to 57%.

To elaborate further, Figure 5 and Figure 6 compare the model performance before and after including the text data. The figures describe the model accuracy in both cases using a confusion matrix. As it appears, including the textual data had a pronounced impact on the model accuracy. In particular, the accuracy of predicting the CHIR category improved by a factor of 7%, while other categories remained largely the same. In general, the confusion matrices show that the model could provide the best accuracy of prediction for the MED category.

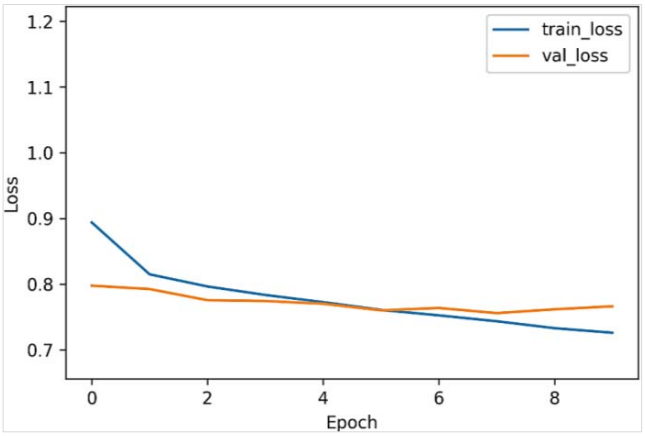


Fig 4. Model loss in train and validation.

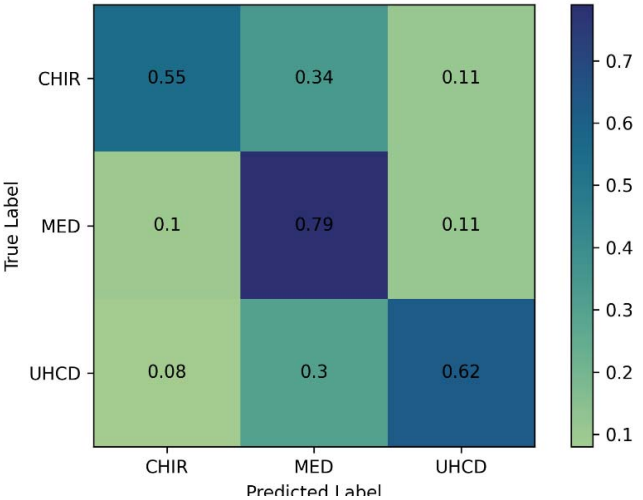


Fig 5. Confusion matrix (without text data).

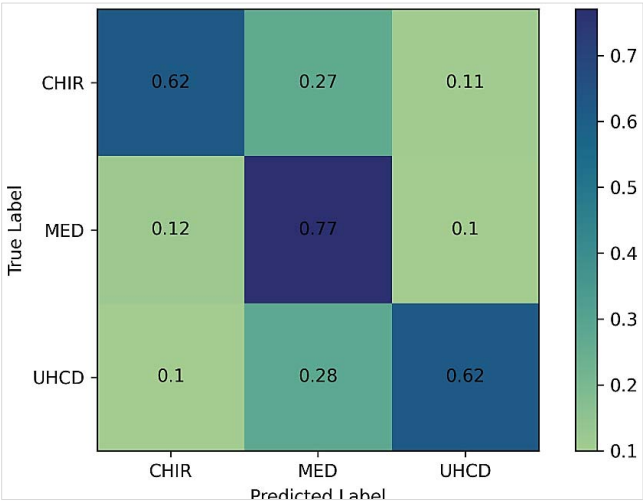


Fig 6. Confusion matrix (after including text data).

V. Conclusions

Several studies aimed to discuss the potentials of NLP for the healthcare domain (e.g. [28-30]). In line with such efforts, this study provided a practical NLP implementation that may support the operations of the ED units. More specifically, we demonstrated that ML-based approaches could be utilized to help provide an early prediction of the medical specialty for hospitalized patients.

The significance of our approach is the effective integration of structured data in tandem with unstructured free-text notes recorded at the triage stage. The conclusion of our empirical results robustly confirmed the positive impact of using such textual notes on the model performance. From a practical standpoint, the study also provides an example of an NLP application using clinical data with languages other than English, which was French in our case.

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In general, the classification model could achieve promising accuracy, which encourages for building a good foundation for future applications with a larger dataset. The models can be deployed to help with the care planning, and improvement of the patient journey from admission to discharge. In this sense, it is intended to experiment the model within real settings in our future work.

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